ZAIN HAMDIA

Engineering & Robotics Student | Builder | Innovator | Volunteer Firefighter

Princeton Day School
Expected Graduation: 2027

FIRE SERVICE

Montgomery Township Volunteer Fire Company #2 | Volunteer Firefighter
July 2025–Present

- Participate in firefighter training, station operations, emergency response activities, and community service as a member of the department's Junior Firefighter Program.
- Leading an initiative to redesign and strengthen the department's Junior Firefighter Program, including updating online information on junior firefighting, streamlining the process for certifying Juniors to respond, and making the mentor-finding process easier between all the firefighters.

ENGINEERING, TECHNOLOGY & RESEARCH EXPERIENCE

Montgomery Township Volunteer Fire Company #2 | Technology & Engineering Intern
Summer 2026–Present

- Redesigned and developed the fire company's public website that transformed the existing site into a modern, accessible resource for department information, membership, community engagement, and recruitment.
- Designing and building a Raspberry Pi-based digital information system for permanent use in the station's radio room, integrating operational information, weather radar and forecasts, emergency alerts, news, and station communications into a centralized dashboard.
- Engineering a custom wall-mounted enclosure for an 11-inch display and Raspberry Pi using CAD and 3D printing, incorporating ventilation, cable management, mounting hardware, and access to power and network connections.
- Developing the system from requirements gathering and design through software/hardware integration, prototyping, reliability testing, installation, firefighter feedback, and technical documentation.

Ivy Tutoring | Software/Technology Intern
Summer 2026–Present

- Develop and enhance features for a startup SAT preparation platform using JavaScript, Java, and C++.
- Built functionality enabling users to log and track SAT practice and test scores over time.

- Developed AI-powered features that analyze prior performance to identify areas for improvement and generate personalized study recommendations.
- Created customized flashcards and practice exercises using the College Board's existing question bank.
- Developed proficiency in Google Antigravity as part of the platform's development and enhancement.

Princeton University Laboratory Learning Program | Summer Research Intern
Summer 2025

- Participated in the Joseph Henry Project research program.
- Conducted scientific research and contributed to academic project work.

ENGINEERING & ROBOTICS PROJECTS

Zainiac Robot Series — Zainiac 19, Zainiac 21 & Zainiac 25
Designed and built multiple original robots incorporating mechanical design, electronics, programming, fabrication, and iterative prototyping.

Little Bot & Noggin
Developed independent robotic projects exploring design, movement, hardware integration, and controls.

3D-Printed Flight Simulator
Designed and constructed a flight simulator incorporating custom-designed and 3D-printed components.

Additional Technical Work

- Self-taught Arduino programming and hardware integration.
- Produced 3D-printed N95 masks donated to a New York cancer center during the COVID-19 pandemic.

Exhibitions & Presentations: Philadelphia Maker Faire • Rochester Maker Faire • Brussels Maker Faire • Cairo Maker Faire • Tandem Industrial Design Conference at Jefferson University

EMPLOYMENT

True Value | Part-Time Associate
July 2026–Present

- Assist customers with product questions, merchandise selection, and general store needs in a customer-facing retail environment.

- Stock and organize new merchandise and support day-to-day store operations while balancing ongoing school, extracurricular, and community commitments.

LEADERSHIP & ACTIVITIES

Princeton Day School

- **Co-President and Team Captain, Robotics Club** | 2025–Present
- **Head of Design Team, Robotics Club** | 2024–2025
  Led design efforts resulting in a 6th-place finish at the February 2025 Robotics Competition.
- **Co-President, Maker Club** | 2024–Present
- **Co-President, Moslem Student Association** | 2024–Present

PUBLICATIONS & RECOGNITION

Published Articles

- *"Virtual Reality, Just What the Doctor Ordered"*
- *"The Clandestine Side of 3D Printing"*
- *"Have We Been Here Before?"*
- *"From COVID-19 to Zainiac 19"*

Recognition

- PSAT National Recognition Program — Rural and Small Town Recognition Award
- Vex V5 Robotics Competition — Quarterfinalist, 2026
- Vex V5 Robotics Competition — 6th Place, 2025
- Featured on Belgian national news following participation in the Brussels Maker Faire and participated in the opening ceremony alongside Brussels government leadership.

TECHNICAL SKILLS

Programming: Python • Java • JavaScript • C++ • Arduino
Design & Engineering: Fusion 360 • CAD • Robotics Design • 3D Printing • Prototyping
Applied Technology: Raspberry Pi • Hardware Integration • Web Development